Results and Observations

The Patient Readmission Prediction model was successfully trained using the Diabetes 130-US Hospitals dataset containing over 100,000 patient encounters. Data preprocessing techniques including missing value handling, feature encoding, and feature selection were applied to prepare the dataset for machine learning.

The XGBoost classifier was trained to identify patients at risk of being readmitted within 30 days of discharge. Model performance was evaluated using standard classification metrics such as Precision, Recall, F1-Score, and ROC-AUC Score to assess predictive capability.

The prediction module successfully generated readmission risk probabilities for individual patient records, classifying patients into High-Risk and Low-Risk categories. This demonstrates the practical application of machine learning in supporting healthcare decision-making and proactive patient management.

Key Achievements

- Successfully processed and analyzed a real-world healthcare dataset with over 100,000 patient records.
- Implemented an end-to-end machine learning pipeline including data preprocessing, model training, evaluation, and prediction.
- Developed a predictive model capable of identifying patients with a higher likelihood of 30-day readmission.
- Utilized XGBoost, one of the most effective algorithms for structured healthcare data.
- Built a reusable prediction framework that can be extended for real-world healthcare applications.

Business Impact

Hospital readmissions are a significant challenge for healthcare providers due to increased treatment costs and resource utilization. Predictive analytics can help healthcare organizations identify at-risk patients before discharge, enabling targeted interventions, follow-up care, and personalized treatment plans.

By leveraging machine learning, this project demonstrates how healthcare data can be transformed into actionable insights that improve patient outcomes, optimize hospital resources, and support evidence-based clinical decision-making.

Future Enhancements

Future versions of this project will focus on:

- Integration with HL7 and FHIR healthcare interoperability standards.
- Explainable AI using SHAP for model transparency.
- Interactive dashboards using Streamlit.
- REST APIs using FastAPI for real-time predictions.
- Cloud deployment using AWS, Azure, or Google Cloud Platform.
- Enhanced feature engineering and advanced ensemble learning techniques.

Conclusion

This project showcases the application of Machine Learning in Healthcare Analytics by predicting patient readmission risk using historical clinical data. It highlights the complete lifecycle of a healthcare AI solution, from data acquisition and preprocessing to model training, evaluation, and deployment readiness. The project serves as a strong foundation for developing advanced predictive healthcare systems and demonstrates the potential of data-driven approaches in improving patient care and operational efficiency.

Output of train.py

```
  df = pd.read_csv(data_url)
Dataset Shape: (101766, 47)

Target Distribution:
readmitted
0    90409
1    11357
Name: count, dtype: int64

Categorical Columns: 33
Numerical Columns: 11

Training...

Classification Report
              precision    recall  f1-score   support

           0       0.89      1.00      0.94     18083
           1       0.59      0.01      0.02      2271

    accuracy                           0.89     20354
   macro avg       0.74      0.50      0.48     20354
weighted avg       0.86      0.89      0.84     20354


Confusion Matrix
[[18070    13]
 [ 2252    19]]

ROC AUC Score: 0.6858

Model Saved Successfully!
```

Output of predict.py

```
c:\Users\arjun\AppData\Local\Programs\Python\Python313\Lib\site-packages\ucimlrepo\fetch.py:97: DtypeWarning: Columns (10) have mixed t
ypes. Specify dtype option on import or set low_memory=False.
  df = pd.read_csv(data_url)
Patient Data:
        race  gender     age  admission_type_id  ...  metformin-rosiglitazone  metformin-pioglitazone  change  diabetesMed
0  Caucasian  Female  [0-10)                  6  ...                       No                      No      No           No

[1 rows x 44 columns]

Prediction Result
Readmission Risk: 5.90%
LOW RISK: Patient unlikely to be readmitted within 30 days
```